

Writing and Solving Linear Inequalities

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

- Not reversing the sign when $\div$ or $\times$ by a negative.
- Reversing when only adding or subtracting.
- $<$ and $\leq$ are not the same: $\leq$ includes the end value.
- Writing $x = \dots$ instead of $x < \dots$

METHOD

- 1 Balance both sides like an equation
- 2 Undo $+/-$ first, then $\times/\div$
- 3 Flip the sign if $\times$ or $\div$ by a negative
- 4 Write the answer as an inequality
- 5 Check with a value from the solution

WORKED EXAMPLE

Solve $3x + 4 < 19$.

Subtract 4: $3x < 15$. Divide by 3: $x < 5$.

Check $x = 4$: $16 < 19$

VISUAL AID

Number line for $x < 5$: an open circle at 5 (5 not included) with the line shaded to the left to show every value less than 5.

1. (2 marks)

Solve: a) $x + 3 > 7$ b) $x - 2 \leq 5$

2. (2 marks)

Solve: a) $2x < 10$ b) $5x \geq 20$

3. (3 marks)

Solve: a) $2x + 1 > 9$ b) $3x - 4 \leq 11$ c) $4x + 5 < 5$

4. (3 marks)

Solve (reverse the sign when $\div$ by a negative): a) $-x > 3$ b) $-2x \geq 8$ c) $5 - x < 2$

5. (3 marks)

Solve: a) $3(x - 2) \geq 9$ b) $2(x + 1) > x + 5$ c) $4x - 7 \leq 2x + 3$

6. (3 marks)

Form and solve: a) Sam has $\pounds x$; after spending $\pounds 12$ has less than $\pounds 20$. b) Taxi $\pounds 4 + \pounds 2/\text{mile}$, cost $\pounds 20$ — most whole miles?

7. (3 marks)

a) Solve $4x + 3 \leq 19$; smallest integer x ? b) Solve $2x - 5 < 7$; largest integer x ?

8. (5 marks)

Rectangle width x , length $(x+4)$, perimeter < 36 cm, x a positive integer. a) Inequality. b) Solve. c) List possible x .